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# NETWORK ANALYSIS REPORT

## Project 3: Login Traffic Analysis

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| **Student Name** | Om Vyankat Chavan |
| **Roll Number** | 093019 |
| **Subject** | Cyber Security / Network Analysis |
| **Target** | http://testasp.vulnweb.com/ |
| **Tools Used** | Wireshark and Burp Suite Community Edition |
| **Test Environment** | Kali Linux 2025.4 (VMware), Firefox 140.0, interface eth0 (192.168.169.128) |
| **Date of Testing** | 08 October 2026 |

## 1. Objective

This practical involved capturing and examining the network traffic produced while logging in to testasp.vulnweb.com, the intentionally vulnerable Acunetix Acuforum demonstration site. The aim was to locate the login request, inspect the captured packets, check whether the login details were protected, and record any security weaknesses found.

## 2. Target and Tools

**Target:** http://testasp.vulnweb.com/ (Acunetix test site, resolved to 44.238.29.244, running Microsoft-IIS/8.5 with ASP.NET)

**Tools:**

- Wireshark – packet capture and protocol analysis on the eth0 interface
- Burp Suite Community Edition (v2025.10.6) – intercepting proxy used to inspect the login request
- Firefox 140.0 on Kali Linux – browser used to submit the login form

## 3. Scope and Ethical Note

The site testasp.vulnweb.com is an Acunetix demonstration website designed for security testing and training. Its own page warns that it is intentionally vulnerable. For this practical, I captured only the login traffic generated from my own virtual machine using the practice credentials lab / lab. I did not monitor anyone else's traffic or carry out an attack against the server.

## 4. Methodology

1. I checked the Kali Linux virtual machine's network settings using ip addr and confirmed that eth0 had the address 192.168.169.128. I then launched Burp Suite.
2. I opened the target website in Firefox and went to the Acuforum login page.
3. I started capturing packets in Wireshark on the active eth0 interface.

4. I entered the practice username and password (lab / lab) and submitted the login form.
5. I used the http display filter to view plain HTTP traffic, then applied http.request.method == "POST" to identify the login submissions.
6. I examined the HTTP POST packet details and checked the raw packet bytes for the submitted form data.
7. I selected Follow → HTTP Stream (tcp.stream eq 17) to review the complete conversation between the client and server.
8. I intercepted the same login request in Burp Suite using Proxy → Intercept to verify its contents.
9. I reviewed the rest of the capture and compared the login traffic with connections protected by TLS encryption.
10. Finally, I compared the results with the security requirement that login information should be sent through an encrypted connection.

## 5. Lab Setup and Login Page

The Kali Linux virtual machine received the address 192.168.169.128/24 on eth0. I launched Burp Suite from the terminal and opened the target login page in Firefox. The page used plain HTTP, which Firefox marked as “Not Secure”; the address bar displayed http://testasp.vulnweb.com/Login.asp.

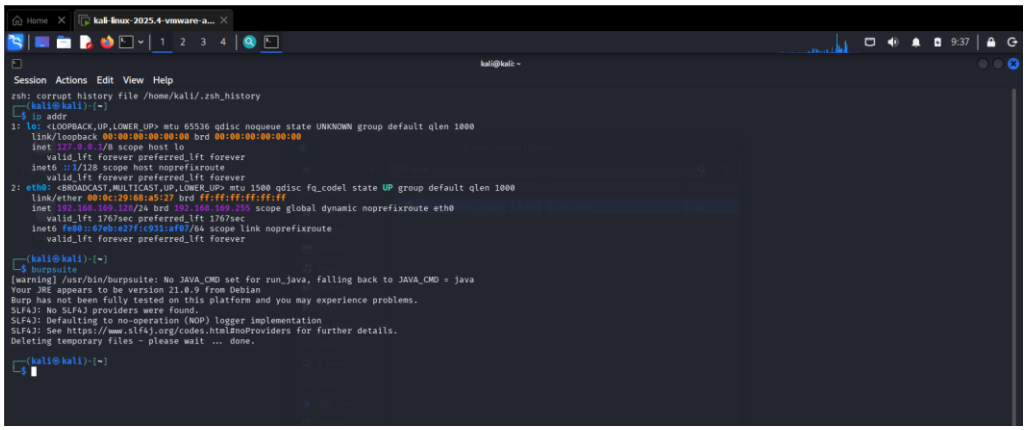



*Figure 1: Kali Linux terminal showing the eth0 address (192.168.169.128) and Burp Suite being launched.*

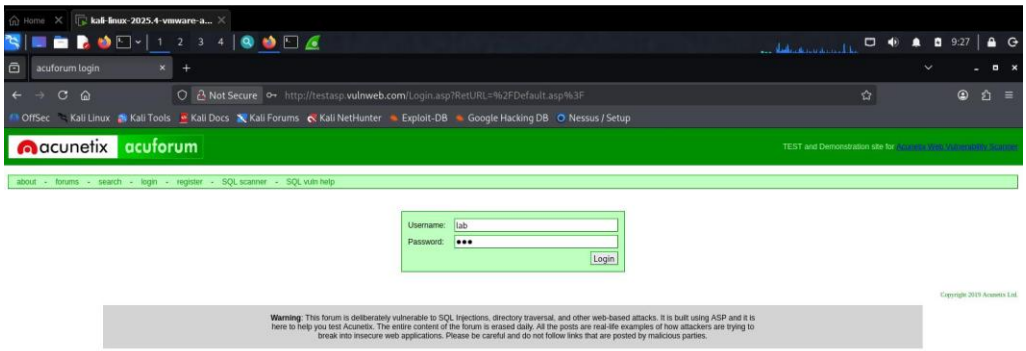



*Figure 2: Acuforum login page opened over HTTP ("Not Secure") with the practice credentials entered.*

# 6. Wireshark Analysis

## 6.1 Identifying the login request

After I applied the http filter, Wireshark showed a GET request for the home page and login page, followed by two POST requests to /Login.asp?RetURL=%2FDefault%2Easp%3F (packets 752 and 767). These requests travelled from 192.168.169.128 to 44.238.29.244. The server returned HTTP/1.1 200 OK (text/html). The captured requests and responses all used plain HTTP over TCP port 80.

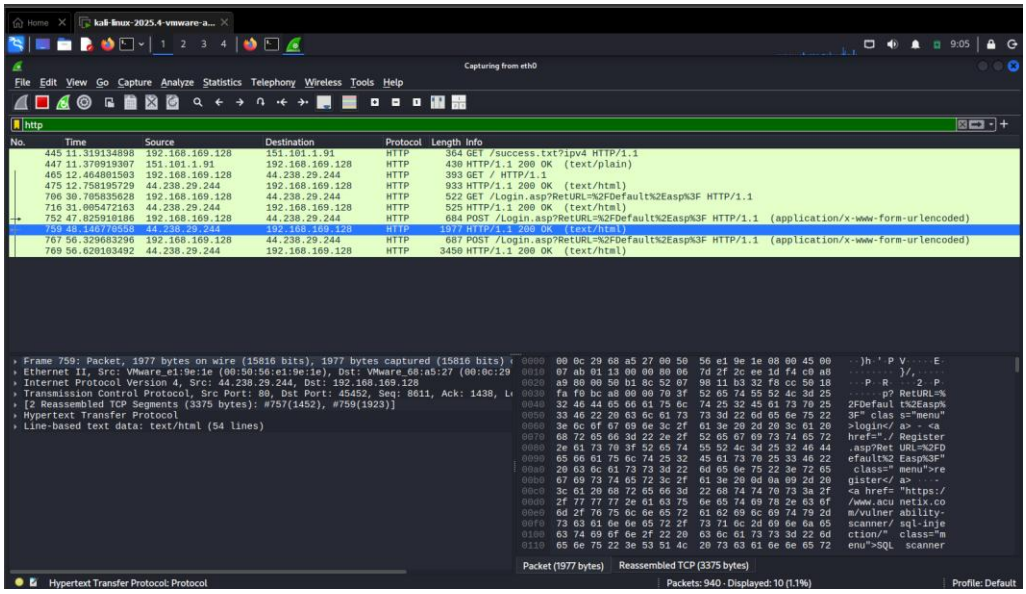



*Figure 3: Wireshark http filter showing the GET requests, the login POST requests and the server's 200 OK responses.*

## 6.2 Inspecting the POST packet

Applying http.request.method == "POST" isolated the two login submissions. Packet 767 (687 bytes) was a TCP segment sent from source port 45452 to destination port 80. Its HTTP payload used application/x-www-form-urlencoded encoding and contained 23 bytes of form data. In the packet-bytes pane, the fields tfUName=lab&tfUPass=lab were readable as plain text, exposing both the username and password. The same packet also showed the session cookie, Origin, and Referer headers.

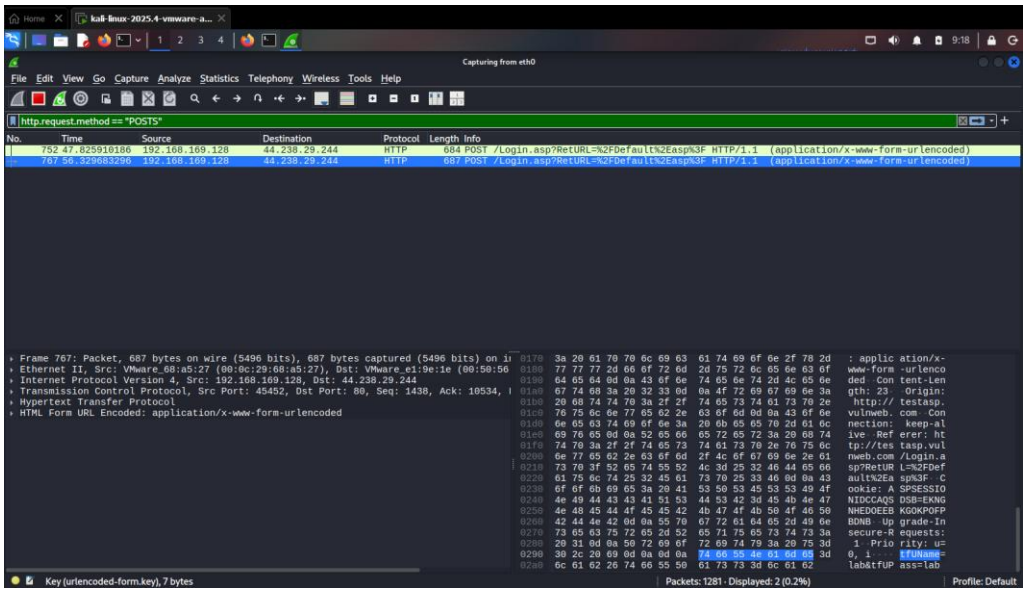



*Figure 4: Wireshark POST filter and packet bytes – username and password readable in plain text.*

## 6.3 Follow HTTP Stream

Using Follow → HTTP Stream on stream 17 displayed the full conversation in readable ASCII: 4 client packets, 4 server packets, and 15 kB of data. The request headers included Host: testasp.vulnweb.com and the Firefox user agent. The server returned HTTP/1.1 200 OK and revealed the headers Server: Microsoft-IIS/8.5 and X-Powered-By: ASP.NET. It also issued an ASPSESSIONID session cookie with path=/ but without the Secure or HttpOnly attributes, followed by the page HTML.

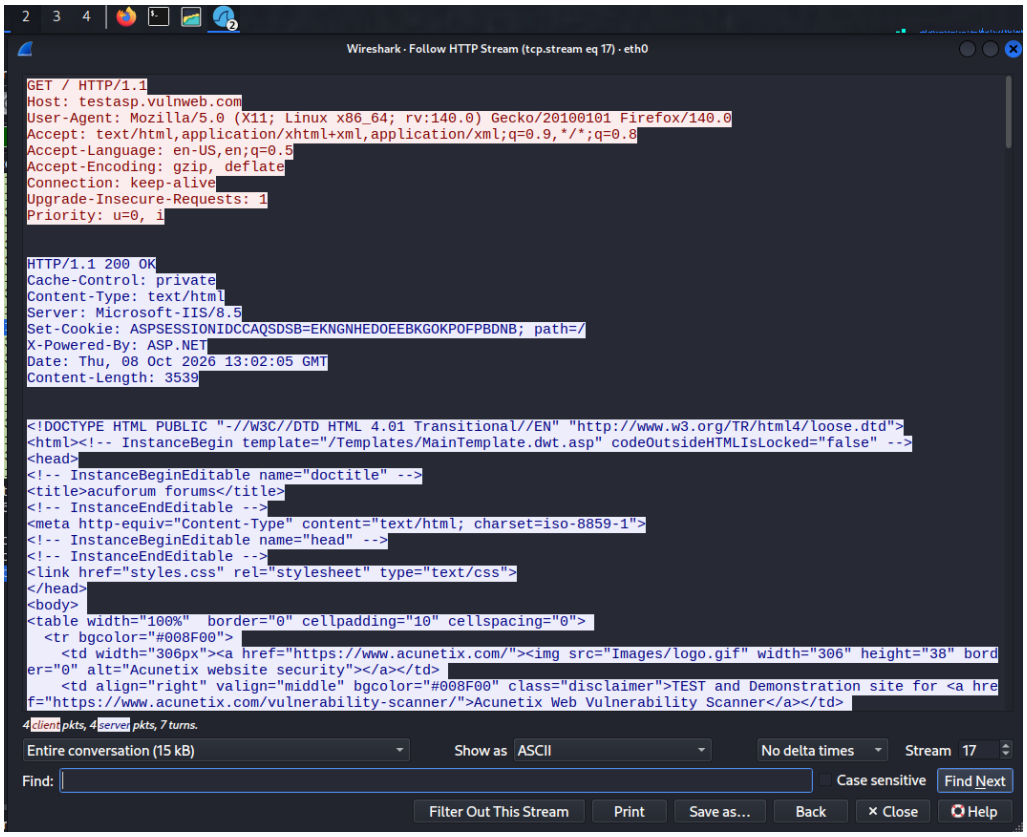



*Figure 5: Wireshark Follow HTTP Stream showing the request, server headers, Set-Cookie header and page content in clear text.*

## 6.4 Comparison with encrypted traffic

Other background connections in the same capture used TLSv1.2 or TLSv1.3 over port 443. Wireshark displayed their contents as “Application Data,” so the data could not be read directly. In contrast, the login traffic to testasp.vulnweb.com was fully readable. This comparison highlights the missing encryption on the login connection.

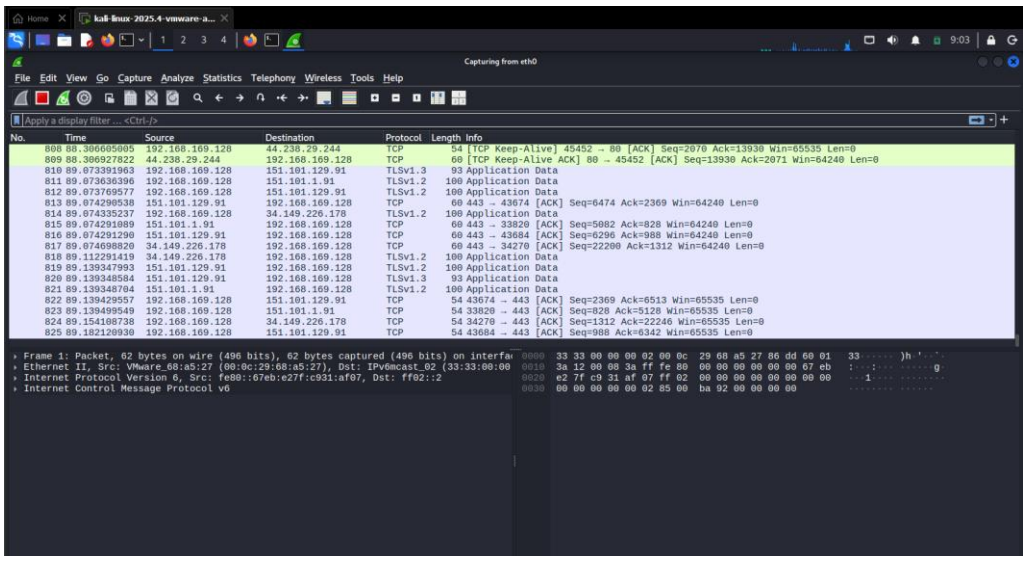



*Figure 6: Unfiltered capture – other connections use TLS (port 443) and show only encrypted “Application Data”.*

## 7. Burp Suite Analysis

I used Burp Suite Community Edition as an intercepting proxy to inspect the login request. Burp displayed "Request to http://testasp.vulnweb.com:80 [44.238.29.244]," confirming that the connection used HTTP port 80. The request was POST /Login.asp?RetURL=%2FDefault%2Easp%3F HTTP/1.1, with Content-Type application/x-www-form-urlencoded and Content-Length 23. The Cookie header contained the ASPSESSIONID session identifier, while the request body contained tfUName=lab&tfUPass=lab. These observations matched the Wireshark results and confirmed that both the login details and session identifier were sent in an unencrypted request that could be read or modified while in transit.

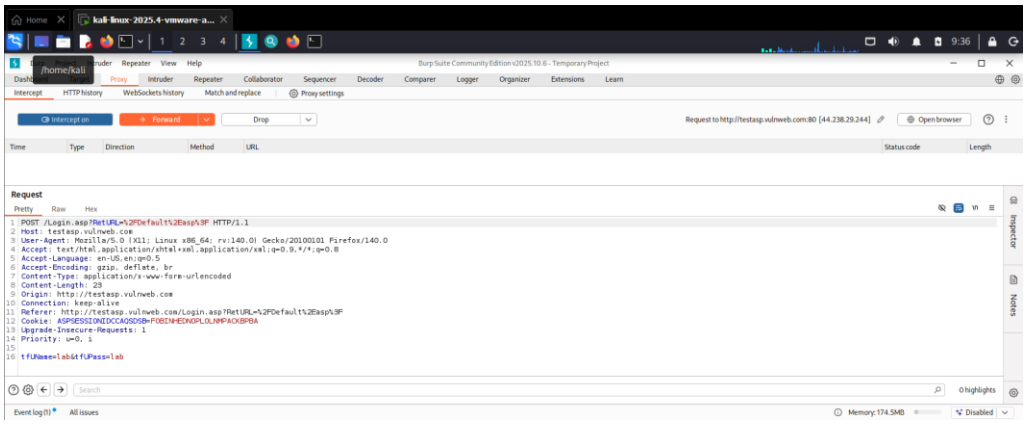



*Figure 7: Burp Suite Proxy → Intercept showing the login POST request, session cookie and credentials in the body.*

## 8. Findings

| Item | Observation | Result |
|---|---|---|
| Protocol | HTTP on TCP port 80 (testasp.vulnweb.com, 44.238.29.244) | Not encrypted |
| Login method | POST /Login.asp?RetURL=%2FDefault%2Easp%3F | Login request captured |
| Form encoding | application/x-www-form-urlencoded (23 bytes) | Readable form data |
| Authentication data | tfUName and tfUPass fields visible in Wireshark and Burp | Sensitive data exposed |
| Session cookie | ASPSESSIONID sent in Cookie header and issued via Set-Cookie with path=/ only | Exposed; no Secure / HttpOnly flag seen |
| Server information | Server: Microsoft-IIS/8.5, X-Powered-By: ASP.NET | Technology disclosed |
| HTTPS/TLS | Not used for the observed login traffic | Security weakness |
| HSTS / HTTPS redirect | No Strict-Transport-Security header in the captured response; page served over HTTP | Not enforced |

## 9. Security Weaknesses

1. Cleartext transmission of credentials. The login form uses unencrypted HTTP, so the username and password can be read from the packets travelling between the client and server. Someone able to monitor the network traffic, such as on shared Wi-Fi or through a compromised router, could see these details. Figures 4 and 7 show the credentials in the captured traffic.

2. Exposed session identifier. The ASPSESSIONID cookie is transmitted in plain text. In the captured response, it was set without the Secure and HttpOnly attributes. If someone obtained this cookie, they could attempt session hijacking; the missing HttpOnly attribute also makes it accessible to browser-side scripts.

3. Man-in-the-middle risk. Because TLS is not used, the connection does not provide encryption, integrity protection, or TLS-based server authentication. The Burp inspection demonstrates that a request could be changed in transit, and a person positioned on the network path could alter or spoof responses.

4. No transport-security enforcement. The captured traffic showed no redirect to HTTPS and no HSTS header. As a result, the user was not directed to a secure connection during this test.

5. Information disclosure. The Server and X-Powered-By headers reveal that the site uses IIS 8.5 and ASP.NET. This information could help an attacker identify relevant, version-specific weaknesses.

## 10. Risk Assessment

| Weakness | Potential impact | Risk level |
|---|---|---|
| Credentials sent over HTTP | Account takeover through network sniffing | High |
| Session cookie without Secure/HttpOnly, sent over HTTP | Session hijacking, cookie theft | High |
| No TLS (man-in-the-middle) | Tampering with requests and responses | Medium–High |
| No HTTPS redirect / HSTS | Users stay on an insecure channel | Medium |
| Server/technology headers exposed | Easier targeted attacks | Low |

Note: the risk levels are the author's qualitative assessment for a real-world site; testasp.vulnweb.com is a demo environment and holds no real user data.

## 11. Screenshots Included

- Figure 1 – Terminal: network interface and Burp Suite launch
- Figure 2 – Acuforum login page over HTTP ("Not Secure")
- Figure 3 – Wireshark http filter: GET, POST and 200 OK packets
- Figure 4 – Wireshark POST filter and packet bytes showing the credentials
- Figure 5 – Follow HTTP Stream (tcp.stream eq 17)
- Figure 6 – Unfiltered Wireshark capture: encrypted TLS traffic for comparison
- Figure 7 – Burp Suite intercepted login request

## 12. Summary

This practical successfully captured and examined the login traffic to testasp.vulnweb.com using Wireshark and Burp Suite. The login request travelled over HTTP on port 80, and the username, password, and session cookie were visible in the packet capture, HTTP stream, and intercepted Burp request. This means the authentication information was not protected in transit. The main issue identified was the lack of HTTPS/TLS, which can expose users to traffic monitoring, session hijacking, and man-in-the-middle attacks. The site should use HTTPS/TLS throughout the login process and authenticated session, along with HSTS and secure cookie settings.

## 13. Suggestions / Recommendations

- Use HTTPS with TLS (version 1.2 or higher) for the login page and for all authenticated sessions.
- Never transmit usernames, passwords, session identifiers or other sensitive data over plain HTTP.
- Redirect all HTTP requests to HTTPS and enable HTTP Strict Transport Security (HSTS).
- Set the Secure and HttpOnly attributes (and a suitable SameSite value) on session cookies.
- Remove or mask the Server and X-Powered-By headers to reduce information disclosure.
- Regenerate the session identifier after login and apply sensible session timeouts.
- Use a valid, trusted certificate and disable outdated protocols and weak cipher suites.